

UNITY GAME DESIGN DOCUMENT

VaultQuest

A 4-Scene Jumanji-Style Adventure Game

Egypt Dunes → Zombie City → Dino Jungle → Medieval Kingdom

ENGINE	Unity 2022 LTS (or Unity 6)
TOTAL SCENES	4 Scenes + Hub (Egypt Sand Dunes)
GAME TYPE	3D Third-Person Adventure / Stealth-Survival
WIN CONDITION	Collect all coins from all scenes, return to Hub
LOSE CONDITION	Caught in final scene with no escape — Game Over
CORE MECHANIC	Dice Roll triggers scene jump — Jumanji-style
TARGET PLATFORM	PC / VR-ready architecture
PROTAGONIST	Arjun — 14-year-old boy, agile & resourceful
AI VIDEO TOOL	NotebookLM (this doc is your upload source)

This document is the complete game design bible for VaultQuest. It covers every scene's gameplay mechanics, Unity implementation notes, C# scene-management logic, coin systems, enemy AI behaviour, visual specifications, and NotebookLM video prompt scripts for AI-generated story video creation.

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What is VaultQuest?

VaultQuest is a Unity-built 3D adventure game inspired directly by the Jumanji board-game mechanic. The player controls Arjun, a 14-year-old boy who discovers a mysterious ancient box in a house hidden among the Egyptian sand dunes. When he rolls the dice inside the box, the game world comes alive and pulls him into dangerous parallel worlds. Each world holds a coin — the only currency for returning home. Collect all coins, return to the hub, and Arjun wins the game and escapes.

The Jumanji Core Loop

STEP	ACTION	UNITY MECHANIC
1	Player arrives at Egypt Hub (Sand Dunes House)	Scene: EgyptHub loads on game start
2	Player finds the ancient box inside the house	Trigger zone activates box UI
3	Player opens box — dice roll animation plays	Dice animation + RNG number (1-6)
4	Dice determines which world opens next	SceneManager.LoadScene() by dice value
5	Player enters the world — must find the coin	NavMesh + collectible coin GameObject
6	Coin found — player can portal back to Hub	Portal activates when coin collected
7	Enemy catches player — sent to next danger world	OnTriggerEnter() fires scene load
8	Player escapes from that world too — returns to Hub	Same portal mechanic per scene
9	All coins collected in Hub — Win condition fires	WinManager checks coin count == 3

Player Character — Arjun

Arjun is a 14-year-old Indian boy — agile, quick-thinking, and slightly reckless. In Unity, he is controlled as a third-person character using Unity's Character Controller component. He has three movement states: Walk, Sprint, and Crouch (crouch is essential for hiding from enemies in all scenes).

ARJUN'S CORE ABILITIES

- Sprint — hold Shift to run faster (increases enemy detection radius slightly)
- Crouch — press C to crouch-hide behind objects (reduces enemy detection to near zero)
- Interact — press E to open box, roll dice, collect coins, activate portals
- Eagle Sense — press Q to highlight nearby coins and enemies (3-second cooldown)
- No combat — Arjun cannot fight enemies; stealth and speed are his only tools



Detailed Flow Logic

The game flow is non-linear by penalty — the player ideally completes Scene 1 (Zombie City), returns to Hub with coin, then rolls dice again for Scene 2 (Jungle), and so on. However, getting caught by enemies creates a forced chain:

SITUATION	FROM	GOES TO	TRIGGER
Normal play — coin found	Zombie City	Egypt Hub	Coin collected + portal activated
Caught by zombie	Zombie City	Dinosaur Jungle	OnTriggerEnter(zombie) fires
Normal play — coin found	Dino Jungle	Egypt Hub	Coin collected + portal activated
Caught by dinosaur	Dino Jungle	Medieval Kingdom	OnTriggerEnter(dino) fires
Normal play — coin found	Medieval Kingdom	Egypt Hub	Coin collected + portal activated
Caught by soldier	Medieval Kingdom	GAME OVER	No further escape scene — game resets
All 3 coins in Hub	Egypt Hub	WIN SCREEN	WinManager.CheckCoins() == 3
Player rolls dice again in Hub	Egypt Hub	Next scene from dice roll	Die RNG picks uncompleted scene

Scene 01 — Egypt — Sand Dunes Hub

UNITY

SCENE
01

Egypt — Sand Dunes Hub

The Starting World | Unity Scene: EgyptHub

Unity Scene

Scene Overview

The Egypt Sand Dunes Hub is the central world of VaultQuest — Arjun always returns here between worlds. It is set in a vast Egyptian desert at golden hour. Sand dunes roll in every direction. Half-buried ruins and obelisks break the horizon. At the centre of the playable area sits an ancient stone house — barely visible from the outside, half-consumed by sand. This is where everything begins.

ENVIRONMENT & SETTING

- Vast sandy terrain using Unity's Terrain system — rolling dunes
- Egyptian stone house at map centre — interior is explorable
- Half-buried obelisks and sphinx fragments scattered as natural ruins
- Distant pyramids on the horizon (non-interactive skybox detail)
- Sky: Deep amber sunset gradient — golden light casting long shadows
- Wind particle system — blowing sand particles using Unity's Particle System
- Interior of house: dark, dusty, low ceiling — single shaft of light from window

GAMEPLAY MECHANICS

- Player spawns outside the house at game start — walks across dunes to reach house
- Interactive door — press E to enter the house (door swing animation)
- Inside: ancient wooden box sits on a stone pedestal in the centre of the room
- Press E to open box — lid-opening animation plays (3 seconds)
- Press E again to roll dice — animated 3D dice rolls across stone floor
- Dice value (1-6): 1-2 = Zombie City, 3-4 = Jungle, 5-6 = Medieval (reroll if 1-2)
- Bright portal door opens in house wall — player walks through to enter world
- Upon returning with a coin: coin floats up to UI coin counter in top-right HUD
- When all 3 coins collected: gold beam of light erupts from box, WIN sequence triggers

Unity Assets Required

ASSETS

- Terrain: Unity Terrain with sand texture (normal + height maps), dune mesh brush
- Egyptian house: Custom low-poly mesh OR asset from Unity Asset Store (Egyptian Architecture pack)
- Ancient box: Custom asset with lid animation (Animator component, boolean trigger)
- Dice: 3D dice prefab with Rigidbody — use Physics simulation for roll, read top face value
- Portal: Particle system (VFX Graph) — swirling sand vortex in amber/gold colours
- Lighting: Directional light at 15-degree angle, amber colour temperature (6000K warm)
- Skybox: Custom gradient (amber top to dark brown horizon) OR HDRI desert sunset
- Coin UI: Canvas overlay with 3 coin icons — fill with gold glow when collected

Unity Implementation Notes

DEVELOPER NOTES

- Scene name in Unity: 'EgyptHub' — always loaded as scene index 0
- Use DontDestroyOnLoad for GameManager to persist coin count across scenes
- Dice roll: Use UnityEngine.Random.Range(1,7) — map value to scene name string
- Portal activation: SetActive(true) on portal GameObject when coin lands
- Coin counter UI: Update via GameManager.Instance.coinCount++ on collection

Scene 02 — Zombie City

UNITY

SCENE
02

Zombie City

Danger World 1 | Unity Scene: ZombieCity

Unity Scene

Scene Overview

The Zombie City is a post-apocalyptic urban environment — abandoned streets, overturned vehicles, crashed helicopters, shattered storefronts, overgrown with vines and rust. The sky is permanently overcast, a sickly yellow-green. Dozens of zombies patrol the streets in patterns. Arjun must navigate through the city, find the glowing coin hidden in one of the buildings, and reach the return portal without being caught. If a zombie touches him — he's instantly transported to the Dinosaur Jungle.

ENVIRONMENT & SETTING

- Urban grid layout — 4 main streets forming a cross, with a grid of smaller streets
- Building facades only (no full interiors except the coin building)
- Overturned cars, rubbish bins, broken lampposts as obstacles
- Coin building: a half-collapsed convenience store with a glowing coin inside
- Return portal: spawns on rooftop of coin building once coin is collected
- Sky: Yellow-green overcast, permanent dusk lighting, flickering clouds
- Ground: Cracked asphalt with grass growing through, puddles of oil
- Ambient: Distant moaning, wind through empty streets, occasional zombie groans

GAMEPLAY MECHANICS

- Arjun spawns at city edge — must navigate to coin building at far end of map
- Zombies patrol in fixed looping paths — visualise patrol route as NavMeshAgent
- Zombies have detection radius (sphere collider trigger) — if Arjun enters it, they start chasing
- Crouching reduces detection radius by 70% — essential for sneaking past zombies
- Sprinting makes noise — increases zombie detection radius by 30% for 2 seconds
- Arjun can hide behind cars, bins, walls — crouched behind obstacle = undetectable
- Coin is inside broken convenience store — enter through broken window (ladder on side)
- Collecting coin triggers portal to appear on store rooftop — ladder on side
- Portal return: Arjun steps through and arrives at Egypt Hub with Zombie City
- If zombie's sphere collider touches Arjun: screen flash red, fade to black, lose health

Enemy Types

ENEMIES & AI BEHAVIOUR

- Zombie type: Slow Wanderers (most common) — follow patrol path, detection radius 8m
- Zombie type: Sprinting Zombie (2-3 per map) — faster, detection radius 12m, triggered by sprint noise
- Zombie type: Stationary Groaner (near coin building) — stands still but has 360-degree wide radius
- All zombies: Green-grey skin, ragged clothing, empty white eyes, shambling walk animation
- Chase state: If Arjun detected, zombie enters NavMeshAgent chase mode — follows player position
- Chase break: Arjun can break line-of-sight by hiding — zombie returns to patrol after 5 seconds

Unity Assets Required

ASSETS

- City kit: Synty Studios Polygon Apocalypse pack (Unity Asset Store) OR custom modular buildings
- Zombie models: Mixamo zombie animations OR Unity Asset Store zombie character pack
- NavMesh: Bake NavMesh for zombie pathfinding — mark patrol waypoints as empty GameObjects
- Detection: SphereCollider on zombie (trigger=true) + OnTriggerStay to detect Player tag
- Coin: Glowing 3D coin prefab — PointLight component (green-gold), slow rotation animation
- Portal (exit): Same vortex VFX as Hub portal but green-tinted to match zombie world
- UI: Alert icon above zombie head when entering detection range (warning), red flash on catch

Unity Implementation Notes

DEVELOPER NOTES

- Scene name: 'ZombieCity' — loaded by `SceneManager.LoadScene('ZombieCity')`
- If caught: `SceneManager.LoadScene('JungleScene')` — no coin earned
- If coin collected + portal used: `GameManager.Instance.AddCoin('zombie'), LoadScene('EgyptHub')`
- Zombie AI: Use NavMeshAgent + simple state machine (Patrol / Alert / Chase / ReturnToPatrol)
- Patrol: Zombie loops through array of waypoint Transforms using `SetDestination()`
- Detection: `OnTriggerEnter` sets state to Chase; `OnTriggerExit` starts 5s coroutine back to Patrol

SCENE
03

Dinosaur Jungle

Danger World 2 | Unity Scene: JungleScene

Unity Scene

Scene Overview

The Dinosaur Jungle is a lush, prehistoric rainforest — towering ferns, ancient trees with canopies that block the sky, rivers, and muddy paths. The world feels enormous and alive. Multiple species of dinosaurs roam the jungle — some herbivores that ignore Arjun, but three types of carnivores that will chase and catch him. The coin is hidden deep in the jungle, inside an ancient stone ruin. Arjun must traverse the jungle using sound and shadows to avoid the carnivores.

ENVIRONMENT & SETTING

- Dense jungle terrain — Unity Terrain with heavy forest us
- Visibility is low — thick canopy blocks most sky light, crea
- Muddy river cuts across middle of map — shallow enough
- Ancient stone ruin at map centre — Mayan-inspired, overg
- Return portal spawns on top of the altar stone once coin c
- Ground: Thick undergrowth, large ferns (can hide Arjun w
- Sky visible only in clearings — deep blue with prehistoric s
- Audio: Constant jungle ambience — insects, distant dinos

GAMEPLAY MECHANICS

- Arjun spawns at jungle edge (where he fell from Zombie City portal)
- Path to ruins is not straight — must navigate around large herbivore herds
- Crouching in tall ferns = invisible to dinosaurs (visual concealment mechan
- Mud near river: Arjun leaves visible footprint marks — carnivore raptors ca
- Stone path avoids mud — safer route but more exposed (no fern cover)
- Coin on altar triggers trap: stone door closes briefly, must solve simple pre
- Puzzle: Step on the 3 mossy stones in the correct order (marked by symbol
- Portal activates on altar roof — Arjun must climb vine on altar side to reach
- If dinosaur touches Arjun: screen shake + roar sound, fade to black, load M

Enemy Types

ENEMIES & AI BEHAVIOUR

- Velociraptor (x4): Fast, small — hunt in pairs. Detection based on sound (sprinting = instant alert). Tracking footprints in mud is secondary
- T-Rex (x1): Massive, slow — vision-based detection. Sight cone 60 degrees forward, range 25m. Cannot fit through dense forest. Stick to
- Pterodactyl (x2): Flying overhead — drop-swoops at Arjun if spotted in open areas. Stay under tree cover to avoid. Not trigger for next sc
- Brachiosaurus (x3): Non-hostile — but walking path can crush Arjun if he stands in their path. Move aside when they approach.

Unity Assets Required

ASSETS

- Jungle terrain: SpeedTree or Realistic Trees & Plants pack (Unity Asset Store)
- Ferns: Dense fern cover in clusters — use LOD system for performance
- Dinosaur models: Jurassic World Evolution-style pack OR custom Mixamo-rigged dinosaurs
- T-Rex: NavMeshAgent with large stopping distance, vision cone using Physics.OverlapSphere with angle check
- Raptor: NavMeshAgent with faster speed (8-12 m/s), pack behaviour using shared alert state
- Sound detection: Use AudioListener + custom SoundDetector script comparing noise level to raptor threshold
- Coin: Same coin prefab but with blue-green point light (jungle tint)
- Pressure plate puzzle: 3x plate GameObjects with Animator (press/release states) + puzzle manager script

Unity Implementation Notes

DEVELOPER NOTES

- Scene name: 'JungleScene' — loaded from ZombieCity (caught) or EgyptHub (dice roll)
- If caught by carnivore: `SceneManager.LoadScene('MedievalScene')`
- If coin + portal: `GameManager.Instance.AddCoin('jungle')`, `LoadScene('EgyptHub')`
- Footprint system: Instantiate footprint decal prefab at player position every 0.5s when on mud layer
- Raptor tracking: Raptor checks if footprint `GameObject` is in range every 2s, alerts if found
- Fern hiding: Fern trigger zone sets `playerHidden=true`; dinosaur AI checks this flag before chasing
- T-Rex vision: Cast `Physics.OverlapSphere` at T-Rex position, filter by angle to forward direction

Scene 04 — Medieval Kingdom

UNITY

SCENE
04

Medieval Kingdom

Danger World 3 | Unity Scene: MedievalScene

Unity Scene

Scene Overview

The Medieval Kingdom is a sprawling stone castle town — cobblestone streets, market stalls, stone walls, turrets, and a grand castle at the centre. It is daytime, busy-feeling despite the danger. Armoured soldiers patrol in formations, doing rounds of the town. Arjun must sneak through the kingdom, find the final coin (hidden in the castle dungeon), and escape through a portal in the castle courtyard. If a soldier catches him — Game Over, no further escape world.

ENVIRONMENT & SETTING

- Medieval town layout — outer wall, main gate, market street
- Castle: Imposing stone structure at map centre — tall tower
- Market street: Stalls, barrels, carts — excellent hiding spots
- Soldier barracks: Avoid at all costs — highest density of soldiers
- Dungeon entrance: Through castle gate, down spiral staircase
- Return portal: Inside dungeon near coin — can only be used once
- Lighting: Bright midday sun, hard shadows from walls and buildings
- Ambient: Distant blacksmith hammering, horse sounds, soldier marching

GAMEPLAY MECHANICS

- Arjun spawns outside the outer town wall — must find a gap in the wall or a gap in the wall
- Gap: A hole in the outer wall near the stables — hidden but findable by following the path
- Market street: Most soldiers here. Crouching behind stalls and barrels is essential
- Castle gate: Two soldier guards stand here — must wait for patrol gap or distract them
- Distraction: Pick up a stone (press E on ground near stall), throw it (right-click)
- Dungeon is dark — Arjun's vision limited without torch. Eagle Sense highlights enemies
- Coin on prisoner bench — just pick up, no puzzle in this scene (hardest navigation)
- Portal activates instantly in dungeon — return is urgent as soldier patrol timer starts
- Soldier catch = instant Game Over (no further scene) — screen goes red, no further scene

Enemy Types

ENEMIES & AI BEHAVIOUR

- Foot Soldier (x8): Patrol in pairs on fixed routes. Detection radius 10m. Armed with spears — can hear sprinting from 20m.
- Guard Captain (x1): Patrols castle inner courtyard alone. Detection radius 18m, very fast movement, alerts all soldiers in area if he spots Arjun
- Tower Archers (x2): Stationed on towers overlooking main street. Sight-based detection from above — very long range (30m) but limited field of view
- All soldiers: Full armour (chainmail or plate), helmet with visor, castle livery colours (red/silver). March in step-cadence animation.

Unity Assets Required

ASSETS

- Medieval town: Synty Studios Polygon Knights pack OR Unity Asset Store medieval city kit
- Soldier models: Unity Humanoid rig with medieval armour — Mixamo marching/idle animations
- Castle interior: Separate interior scene area loaded additively OR same scene with camera zone triggers
- Dungeon: Dark stone corridor — point lights from wall torches (flickering light animation)
- Coin: Warm gold point light — brightest/most dramatic coin reveal of all scenes
- Distraction stone: Rigidbody sphere (small) — throw using Rigidbody.AddForce in camera forward direction
- Sound detection: Soldiers use same SoundDetector script — sprint noise triggers alert in 20m radius
- Game Over screen: Red vignette overlay + 'CAPTURED!' text Canvas + Restart button

Unity Implementation Notes

DEVELOPER NOTES

- Scene name: 'MedievalScene' — final danger scene. No further scene if caught.
- If caught: Do NOT load another scene. Show GameOver Canvas, offer RestartGame() which loads EgyptHub fresh
- RestartGame(): Call GameManager.Instance.ResetAll() then SceneManager.LoadScene('EgyptHub')
- If coin + portal: GameManager.Instance.AddCoin('medieval'), LoadScene('EgyptHub')
- Stone throw: Instantiate stone prefab, apply AddForce, stone GameObject has trigger — soldiers react
- Soldier alert system: SoldierAlertManager (singleton) — if Captain detects player, set globalAlert=true, all soldiers enter chase
- Dungeon additive load: Use LoadSceneMode.Additive for dungeon sub-area to avoid full scene reload

The Coin System

Coins are the core reward mechanic of VaultQuest. There are exactly 3 coins in the game — one per danger world. Each coin is a glowing 3D collectible with a world-specific colour tint. All coins are managed by the GameManager singleton which persists across scene loads using DontDestroyOnLoad.

COIN	WORLD	LOCATION	COLOUR	CONDITION TO COLLECT
Coin 1	Zombie City	Inside convenience store	Green-gold glow	Enter store, approach coin, press E
Coin 2	Dino Jungle	On altar in stone ruins	Blue-green glow	Collect coin, then solve 3-plate puzzle to open exit
Coin 3	Medieval Kingdom	Dungeon prisoner bench	Warm gold glow	Enter dungeon, approach bench, press E

Win Condition

Win fires when `GameManager.coinCount == 3` AND player is in `EgyptHub`. When Arjun brings the third coin back to the hub, a golden beam of light erupts from the ancient box. The box flies open. A WIN SCREEN appears: 'ARJUN ESCAPES! The ancient game is complete.' — with a replay and main menu button.

Lose Condition

Lose fires ONLY in the Medieval Scene. If a soldier catches Arjun here, a 'CAPTURED!' Game Over screen appears with options to restart from `EgyptHub` (coins reset) or quit. There is no partial save — the game resets fully on loss.

Penalty Mechanic (Jumanji-style)

Being caught in Zombie City or Dino Jungle is not a Game Over — it is a penalty that makes the journey harder by sending Arjun to the next, more dangerous world without a coin. This means the player must navigate MORE worlds to earn all 3 coins. A skilled player might clear all 3 scenes from the hub rolls without ever being caught. A punished player might chain all 3 danger worlds in a row before returning.

GameManager.cs — Singleton with DontDestroyOnLoad

```
// GameManager.cs
using UnityEngine;
using UnityEngine.SceneManagement;

public class GameManager : MonoBehaviour {
    public static GameManager Instance;
    public int coinCount = 0;
    public bool[] coinsCollected = new bool[3]; // 0=zombie,1=jungle,2=medieval

    void Awake() {
        if (Instance == null) { Instance = this; DontDestroyOnLoad(gameObject); }
        else { Destroy(gameObject); }
    }

    public void AddCoin(string world) {
        int idx = world=="zombie"?0:world=="jungle"?1:2;
        if (!coinsCollected[idx]) { coinsCollected[idx]=true; coinCount++; }
        CheckWin();
    }

    void CheckWin() {
        if (coinCount >= 3) { SceneManager.LoadScene("WinScene"); }
    }

    public void ResetAll() {
        coinCount=0; coinsCollected=new bool[3];
        SceneManager.LoadScene("EgyptHub");
    }
}
```

DiceRoller.cs — Dice Roll & Scene Selection

// DiceRoller.cs

```
using UnityEngine;
using UnityEngine.SceneManagement;
using System.Collections;

public class DiceRoller : MonoBehaviour {
    public GameObject portal;
    public Animator diceAnimator;
    private string[] scenes = {"ZombieCity", "ZombieCity", "JungleScene",
                              "JungleScene", "MedievalScene", "MedievalScene"};

    public void RollDice() {
        int roll = Random.Range(0, 6);
        string targetScene = GetAvailableScene(scenes[roll]);
        StartCoroutine(DiceAnimation(targetScene));
    }

    string GetAvailableScene(string preferred) {
        // If preferred scene coin already collected, pick next uncollected
        bool[] c = GameManager.Instance.coinsCollected;
        if (preferred=="ZombieCity" && c[0]) preferred="JungleScene";
        if (preferred=="JungleScene" && c[1]) preferred="MedievalScene";
        if (preferred=="MedievalScene" && c[2]) preferred="WinScene";
        return preferred;
    }

    IEnumerator DiceAnimation(string scene) {
        diceAnimator.SetTrigger("Roll");
        yield return new WaitForSeconds(2f); // dice roll duration
        portal.SetActive(true);
        // Portal stays active -- player walks through to load scene
    }
}
```

EnemyCatch.cs — Universal Enemy Catch Handler

// EnemyCatch.cs — attach to each enemy

```
using UnityEngine;
using UnityEngine.SceneManagement;
using System.Collections;

public class EnemyCatch : MonoBehaviour {
    public string currentScene; // "ZombieCity", "JungleScene", "MedievalScene"
    public float catchCooldown = 1f;
    private bool caught = false;

    void OnTriggerEnter(Collider other) {
        if (other.CompareTag("Player") && !caught) {
            caught = true;
            StartCoroutine(CatchSequence());
        }
    }

    IEnumerator CatchSequence() {
        // Flash screen red here (UI CanvasGroup fade)
        yield return new WaitForSeconds(catchCooldown);
        if (currentScene == "ZombieCity") SceneManager.LoadScene("JungleScene");
        else if (currentScene == "JungleScene") SceneManager.LoadScene("MedievalScene");
        else if (currentScene == "MedievalScene") SceneManager.LoadScene("GameOver");
    }
}
```

Enemy AI Behaviour Specifications

ENEMY	SCENE	SPEED	DETECT RADIUS	DETECT TYPE	CHASE SPEED	STATE MACHINE
Slow Zombie	Zombie City	1.2 m/s	8 m	Proximity sphere	2.5 m/s	Patrol > Alert > Chase > Return
Sprint Zombie	Zombie City	2.0 m/s	12 m	Proximity + sound	5.0 m/s	Idle > Alert > Sprint-Chase
Groaner (guard)	Zombie City	0 m/s	14 m, 360 deg	Proximity always-on	N/A (stationary)	Stationary > Alarm
Velociraptor	Dino Jungle	6.0 m/s	15 m	Sound + footprint	9.0 m/s	Patrol > Track > Chase
T-Rex	Dino Jungle	3.0 m/s	25 m	Vision cone 60 deg	4.5 m/s	Wander > Spot > Charge
Pterodactyl	Dino Jungle	8.0 m/s	30 m (aerial)	Sight (open areas)	10 m/s	Circle > Dive > Return
Foot Soldier	Medieval	1.8 m/s	10 m	Proximity + sound	4.0 m/s	Patrol > Alert > Chase > Alarm
Guard Captain	Medieval	2.5 m/s	18 m	Proximity + vision	5.5 m/s	Patrol > Spot > Alarm ALL
Tower Archer	Medieval	0 m/s	30 m (sight only)	Vision ray 45 deg	N/A	Watch > Alert > Signal

Detection System Design

All enemies share a base detection system but with different parameters. The two detection types are: Proximity Detection (sphere collider trigger — fires when player enters radius regardless of direction) and Vision Detection (Physics raycast or `OverlapSphere` with angle filter — only fires when player is in front of enemy within a cone angle). Stealth override: if `playerHidden` flag is true (player crouched behind cover), detection is suppressed regardless of proximity.

Graphics & Asset Specifications

UNITY

SCENE	RENDER STYLE	SKY	LIGHTING	PALETTE	KEY VFX
Egypt Hub	Low-poly stylised, warm	Desert sunset HDRI	Directional amber, long shadows	Orange, sandy	Glowing sand particles, heat shimmer
Zombie City	Gritty realistic, desaturated	Yellow-green overcast	Green-tinted ambient, flickering lights	Grey, rusty	Vignette, dust motes, zombie breath fog
Dino Jungle	Lush painterly, saturated	Deep blue through canopy	Dappled light, god-rays	Deep jungle green, blue	Sky particles, footprint decals, mist
Medieval	Clean realistic, high contrast	Bright midday blue	Hard white directional, sharp shadows	Grey, red	Torchlight in dungeon, dust in shafts
Portals	All scenes	N/A — world-specific	Portal glow only	Amber (hub), Green (Zombie), Blue (Jungle), Gold (Medieval)	Swirling vortex (portal), Glowing particles

UNIVERSAL GRAPHICS SETTINGS (UNITY URP)

- Render Pipeline: Universal Render Pipeline (URP) — performance-optimised for all platforms
- Post-Processing: Bloom (subtle), Color Grading (per-scene LUT), Vignette (edges only)
- Anti-Aliasing: MSAA x4 or FXAA — quality vs performance choice per platform
- Shadows: Soft shadows, distance 50m — reduce to 30m if performance drops
- LOD: All major assets (trees, buildings, enemies) must have 3 LOD levels
- Draw Distance: Fog used in jungle (green tint, 40m) and zombie city (yellow, 50m)
- Character: Arjun uses standard URP Lit shader — subsurface scattering for skin (optional)

NotebookLM Video Prompt Scripts

Copy these prompts exactly into NotebookLM's video generation module. Each prompt is written to generate one scene's worth of video footage (15-25 seconds per scene). Use the transition prompt between each scene.

OPENING — EGYPT HUB

NOTEBOOKLM PROMPT — OPENING — EGYPT HUB

- Generate a cinematic 15-second clip of a stone house half-buried in Egyptian sand dunes at golden sunset.
- A 14-year-old Indian boy named Arjun (grey hoodie, copper bracelet on wrist) walks across the dunes toward the house.
- Camera: Wide establishing shot showing scale of dunes, then cuts to medium shot of Arjun pushing open the stone door.
- Interior: Dark, dusty room. A single shaft of amber light hits a carved wooden box on a stone pedestal.
- Arjun approaches, opens the box — inside: a set of glowing ancient dice. He picks one up.
- His copper bracelet begins to glow. A portal of swirling amber light opens in the wall beside him.
- Style: Photorealistic, cinematic UE5-quality. Warm amber colour grade throughout.

DICE ROLL TRANSITION

NOTEBOOKLM PROMPT — DICE ROLL TRANSITION

- Generate a 4-second visual: Arjun throws an ancient glowing dice — it tumbles in slow motion.
- As the dice rolls, the room around it dissolves into golden particles.
- The dice lands — a number glows on its face. A bright flash of light.
- Smash cut to: new world environment (specify which world follows this transition).

SCENE 2 — ZOMBIE CITY

NOTEBOOKLM PROMPT — SCENE 2 — ZOMBIE CITY

- Generate a 20-second clip: post-apocalyptic city street at dusk, yellow-green sky, overturned cars.
- Arjun crouches behind an overturned car. Three slow-moving zombies (grey-green skin, white eyes, ragged clothes) shuffle down the street.
- Arjun waits for a gap, then sprints across the road — camera tracks him at low angle.
- He dives through a broken shop window into a convenience store interior.
- Inside: A glowing green-gold coin sits on the dusty counter, pulsing with light.
- He grabs it — instant alert sound. A zombie turns its head toward the store.
- Arjun looks up — stairs visible. Cut to him climbing. Cut to rooftop portal (green vortex).
- Style: Desaturated, gritty realism. Yellow-green ambient light. Slow-motion on sprint moment.

SCENE 3 — DINOSAUR JUNGLE

NOTEBOOKLM PROMPT — SCENE 3 — DINOSAUR JUNGLE

- Generate a 20-second clip: dense prehistoric jungle, massive trees, bioluminescent ferns on ground level.
- Arjun moves carefully through the undergrowth, crouched, parting ferns with his hands.
- A velociraptor (real-scale, feathered, blue-green colouring) steps into view on the stone path ahead — sniffing the air.
- Arjun freezes completely. Hold on his face — wide-eyed, barely breathing.
- Raptor looks away. Arjun exhales and continues — crawling now.
- Wide shot: Stone ruins visible through the jungle ahead, coin glowing blue-green on altar.
- Arjun reaches the altar, grabs the coin. Stone door slams shut — trapped.
- He spots 3 pressure plates. Steps on them in sequence. Door reopens. Vine on altar wall. He climbs.
- Style: Lush green saturation, dappled light, heavy atmosphere. Slow-burn tension.

SCENE 4 — MEDIEVAL KINGDOM

NOTEBOOKLM PROMPT — SCENE 4 — MEDIEVAL KINGDOM

- Generate a 20-second clip: medieval stone town, midday. Cobblestone market street with stalls.
- Arjun peeks around a wall corner — two armoured soldiers in red and silver march past in lockstep.
- He throws a stone down a side alley — it clatters. Both soldiers turn and investigate.
- Arjun darts across the street into a castle gate archway.
- Interior: Dark spiral stone staircase descending to dungeon. He runs down — torch light flickering.
- Dungeon: Stone corridor. On a wooden bench: glowing warm-gold coin.
- He grabs it. Instant golden portal opens on the dungeon wall.
- Arjun steps through — cut to amber sand dunes. He is back. Smiling. Three coins in hand.
- Style: Hard daylight, sharp shadows, warm stone tones. Dramatic dungeon torch flicker.

WIN SEQUENCE

NOTEBOOKLM PROMPT — WIN SEQUENCE

- Generate a 12-second clip: Arjun stands in the stone house interior. Three coins float above his open palm, spinning.
- He turns to the ancient box on the pedestal. Walks toward it slowly.
- Places all three coins inside. The box shakes. Glows. Lid bursts open — a golden beam shoots upward through the cracked roof.
- The beam pierces the desert sky. Sand dunes glow. The house shakes gently.
- Arjun smiles — eyes wide, copper bracelet glowing bright gold.
- Fade to white. Title card: 'VAULTQUEST — YOU ESCAPED.' in bold teal text on dark background.
- Style: Peak emotional resolution — warm gold everywhere. Gentle slow motion. Triumphant.

Recommended Folder Structure

// Unity Project Folder Layout

```
Assets/  
  Scenes/  
    EgyptHub.unity  
    ZombieCity.unity  
    JungleScene.unity  
    MedievalScene.unity  
    WinScene.unity  
    GameOver.unity  
  Scripts/  
    GameManager.cs  
    DiceRoller.cs  
    EnemyCatch.cs  
    EnemyAI.cs  
    CoinCollect.cs  
    PortalTrigger.cs  
    PlayerController.cs  
    SoundDetector.cs  
    FootprintSystem.cs // Jungle only  
    DistractionThrow.cs // Medieval only  
  Prefabs/  
    Characters/ (Arjun, Zombie, Raptor, TRex, Soldier)  
    Environments/ (EgyptHouse, ZombieCity_Kit, JungleKit, MedievalKit)  
    Collectibles/ (Coin_Zombie, Coin_Jungle, Coin_Medieval)  
    VFX/ (Portal, DustParticles, SandWind, TorchFlicker)  
    UI/ (HUD_Canvas, CoinCounter, GameOver_Canvas, WinScreen)  
  Audio/  
    Music/ (ambience per scene)  
    SFX/ (dice, coin collect, zombie moan, roar, soldier march)  
  Materials/ (one folder per scene)  
  Textures/ (one folder per scene)
```

Build Settings — Scene Order

In Unity Build Settings (File > Build Settings), scenes must be added in this exact order for SceneManager.LoadScene() by index to work correctly:

// Build Settings Scene Order

```
Index 0 — EgyptHub  
Index 1 — ZombieCity  
Index 2 — JungleScene  
Index 3 — MedievalScene  
Index 4 — WinScene  
Index 5 — GameOver
```

Key Unity Packages Required

REQUIRED UNITY PACKAGES (PACKAGE MANAGER)

- Universal Render Pipeline (URP) — main render pipeline
- Cinemachine — camera system for follow cam, cutscenes, dice-roll close-up
- Input System (new) — cross-platform player input handling
- NavMesh Components — enemy pathfinding (install from GitHub or Package Manager)
- Visual Effect Graph — portal vortex and particle systems
- TextMeshPro — all UI text (coin counter, prompts, game over screen)
- ProBuilder — optional: quick level blockout before importing final assets
- Post Processing Stack — bloom, colour grading, vignette per scene

END OF DOCUMENT — VaultQuest Unity Game Design Document | v1.0 | 2026

For NotebookLM Video Generation: Upload this entire PDF as the source document.